

2021-2022 学年第一学期期末考试试卷

一、判断题 (10 题)

R1-1 Given a max-heap with unique keys, there are at least $d-1$ keys larger than any key in level d (the root level is 1). ()

R1-2 Suppose that an array is used to store a circular queue, the value of front must be less than or equal to the value of rear . ()

R1-3 { 5, 2, 12, 28, 16, 32, 72, 60 } cannot be the result after the second run of quick sort (assuming the pivot is chosen randomly). ()

R1-4 Quadratic probing isn't equivalent to double hashing with a secondary hash function of $\text{Hash2}(k)=k$. ()

R1-5 Kruskal's minimum spanning tree algorithm implemented by disjoint set with union-by-rank strategy has $O(|E|\log|E|)$ time complexity. Further optimization by introducing path compression improves it to $O(|E|\alpha(|E|, |V|))$ where α is the functional inverse of Ackermann's function. ()

R1-6 Suppose we are now sorting an initial array (8, 3, 9, 11, 2, 1, 4, 7, 5, 10, 6) using Shellsort. If the sorting result after the first run is (1, 3, 7, 5, 2, 6, 4, 9, 11, 10, 8) and that after the second run is (1, 2, 6, 4, 3, 7, 5, 8, 11, 10, 9), then the increments used for the two sortings are 5 and 2. ()

R1-7 Let n be a non-negative integer representing the size of input. The time complexity of the following piece of code is $O(n)$. ()

```
int func(int n)
{
    int sum = 0;
    for (int i=n; i>0; i/=2)
        for (int j=0; j<i; j++)
            sum++;
    return sum;
}
```

R1-8 In an undirected acyclic graph G with 4 connected components, the number of edges will be 4 less than the number of vertices. ()

R1-9 In a tree of degree 4, we have $n_2+n_3+n_4 < n_0$, where n_i is the number of degree i nodes for $0 \leq i \leq 4$. ()

R1-10 The FirstChild-NextSibling representation of a tree is unique. ()

二、单选题 (20 题)

R2-1 When Dijkstra's algorithm is used to find the shortest paths from v_1 to every vertices in an undirected graph G , a distance set $\text{dist}[v]$ is maintained for every vertex v , as the shortest distance from v_1 to v , passing through only the vertices whose shortest path to v_1 has been determined.

Suppose that the destinations are obtained in the order of $v_2, v_3, \dots, v_n$. if there is a vertex w in G so that $\text{dist}[w]$ is decreased during every iteration (except the last one) of the algorithm, how many statements of the following are correct?

(1) w is v_n

(2) w is adjacent to every other vertices

(3) (v_1, w) (if it exists) is the longest edge among all the edges that are adjacent to w .

- A. 2 B. 1 C. 3 D. 0

R2-2 If a complete binary tree with 2000 nodes is represented by an array (root index is 1), then the index of the lowest (deepest) common ancestor of the node at index 910 and the node at index 480 is.

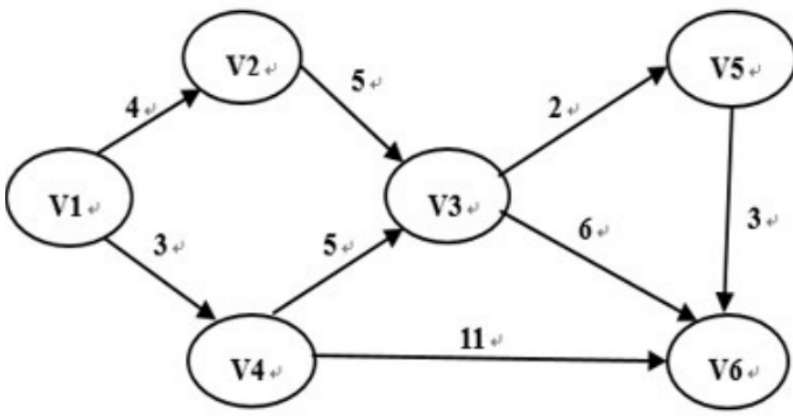
- A. 56 B. 7 C. 14 D. 30

R2-3 To obtain an ascending sequence, which of the following sorting methods gives $\{16, 17, 20, 7, 8, 10, 28, 5, 3\}$ after 2 runs?

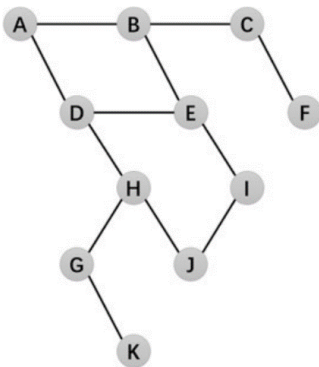
- A. insertion sort B. merge sort
C. selection sort D. quick sort

R2-4 For the given directed weighted graph of 6 vertices, which one of the following sets of vertices satisfies that the earliest completion time of every member is equal to its latest completion time?

- A. $V_1 V_4 V_3 V_6$ B. $V_1 V_2 V_3 V_5 V_6$
C. $V_1 V_4 V_6$ D. $V_1 V_2 V_3 V_6$



R2-5 How many articulation points are there in the undirected graph below?

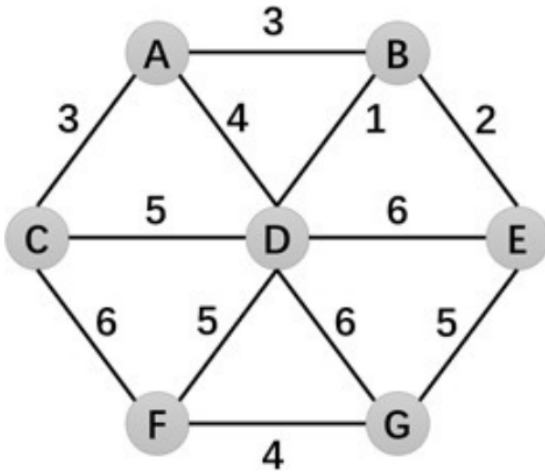


- A. 3 B. 4 C. 2 D. 5

R2-6 Let us convert a general tree T into a binary tree BT . Suppose that there are n_0 leaf nodes in T and m_0 leaf nodes in BT . Which of the following relationship between n_0 and m_0 is true?

- A. $n_0 = m_0$ B. $m_0 \leq n_0$ C. $n_0 \leq m_0$ D. cannot be determined

algorithm?



- A. Starting from B: (B,D) $\rightarrow$ (B,E) $\rightarrow$ (A,B) $\rightarrow$ (A,C) $\rightarrow$ (F,G) $\rightarrow$ (D,F)
- B. Starting from B: (B,D) $\rightarrow$ (A,D) $\rightarrow$ (A,B) $\rightarrow$ (B,E) $\rightarrow$ (E,G) $\rightarrow$ (F,G)
- C. Starting from A: (A,C) $\rightarrow$ (C,D) $\rightarrow$ (D,F) $\rightarrow$ (F,G) $\rightarrow$ (E,G) $\rightarrow$ (B,E)
- D. Starting from A: (A,B) $\rightarrow$ (B,D) $\rightarrow$ (B,E) $\rightarrow$ (A,C) $\rightarrow$ (D,F) $\rightarrow$ (F,G)

R2-11 Which one of the following statement is FALSE about a tree?

R2-12 Given a hash table of size 13 and the hash function $h(x)=x\%13$. double hashing is used to solve collisions with $h(x)=(h(x)+ih_2(x))\%13$ for $i=1,2,\dots,12$, where $h_2(x)=(x\%11)+1$. After filling in the hash table one by one with input sequence $\{92,81,58,21,57,45,161,38,117\}$, which number is placed in the position of index 6?

- A. 57 B. 21 C. 81 D. 58

R2-13 Insert $\{18,23,4,26,50,33,17,39\}$ one by one into a hash table of size 13 with the hash function $H(x)=x\%13$, and linear probing is used to resolve collisions. What is the loading density when the first collision occurs?

- A. 0.31 B. 0.38 C. 0.46 D. 0.23

R2-14 Let n be a non-negative integer representing the size of input. The time complexity of the following piece of code is:

```
Void func(int n)
{
    int i=0;
    while(i*i*i<=n)
        i++;
}
```

- E、 $O(n)$ B、 $O(\log_3 n)$ C、 $O(n^3)$ D、 $O(n^{1/3})$

R2-15 Given a disjoint set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ with array representation. $S[i]$ is initialized to be -1, hence the height of root is 1. If union-by-rank (union-by-height and find-with-path-compression) is used, please list the resulting array after invoking Union(1,2), Union(3,4), Union(1,3), Union(3,5), Union(6,7), Union(1,7), Union(7,8), and Union(4,10).

- A. $\{-2, 1, 1, 1, 1, 1, 1, 1, -1, 1\}$
- B. $\{-3, 1, 1, 3, 1, -2, 6, -1, -1, -1\}$

C. $\{-2, 1, 1, 3, 1, 1, 1, -1, 1\}$

D. $\{-3, 1, 1, 3, 1, 1, 1, -1, -1\}$

R2-16 Push 4 character oppo onto a stack. In how many different ways that we can pop there characters and still obtain oppo?

A. 3

B. 5

C. 6

D. 4

R2-17 A "full tree" of degree 3 (every non-leaf node has 3 children) has 61 nodes. Then its height h is at most . Note: $h=0$ for a single node tree.

A. 20

B. 21

C. 19

D. cannot be determined

R2-18 A binary tree with 2825 nodes must have a height . (Assume the height of a single node tree is 0.)

A. in the range $[12, 1412]$

B. in the range $[10, 2824]$

C. in the range $[11, 2825]$

D. in the range $[11, 2824]$

R2-19 Which of the following is the sufficient and necessary condition that an undirected connected graph has Euler circuit?

A. At most 1 vertex has an odd degree

B. All the vertices have even degrees

C. At most 2 vertices have odd degrees

D. Exactly 2 vertices have odd degrees

R2-20 For a given binary tree, if its post-order traversal sequence is $\{ 5, 2, 6, 4, 1, 7, 3 \}$, pre-order traversal sequence is $\{ 3, 1, 5, 4, 2, 6, 7 \}$, then its in-order traversal sequence must be:

A. 1, 5, 4, 2, 6, 3, 7

B. 5, 1, 2, 4, 6, 3, 7

C. 2, 5, 6, 1, 4, 3, 7

D. cannot be determined

三、程序填空题

R5-1 The following program implements the topological sort algorithm with a stack. Please fill in the blanks.

```

void Topsort(int a[NUM][NUM], int TopNum[NUM])
// a[NUM][NUM] is adjacency matrix of the graph with NUM vertices
// TopNum[NUM] stores the topological orders
{
int S[NUM], Indegree[NUM]; //S[NUM] is a stack
int Counter = 0, top, n, i, j; int V;
top= -1; n=NUM;
for (j=0; j<n; j++)
{ Indegree[j]=0;
  for (i=0; i<n; i++)
    if ( _____ ) Indegree[j]++;
  if ( Indegree[j] == 0 )S[++top]=j;
}
while (top>=0)
{ V = S[top--];
  TopNum[ V ] = ++ Counter; /* assign next */
  for (j=0; j<n; j++)
    if ( a[V][j]!=0)
      if ( _____ == 0 ) S[++top]=j;
} /* end-while */
}

```

R5-2 The function is to sort the array A in descending order.The function Swap(&A[0],&A[i]) is to exchange A[0] and A[i].Please complete the following program.

```

1  #define leftchild(i) (2*(i)+1)
2  void PercDown(ElementType A[], int i, int N) {
3      int child;
4      ElementType tmp;
5      for (tmp = A[i]; leftchild(i) < N; i = child) {
6          child = leftchild(i);
7          if ( _____ )
8              child++;
9          if (tmp > A[child])
10             A[i] = A[child];
11         else
12             break;
13     }
14     A[i] = tmp;
15 }
16 void Heapsort(ElementType A[], int N) {
17     int i;
18     for (i = N / 2; i >= 0; i--)
19         PercDown(A, i, N);

```

```
20     for (i = N - 1; i > 0; i--) {
21         Swap(&A[0], &A[i]);
22         _____;
23     }
24 }
```

四、fn 函数体 R6-1 Print the left subtree of X in BST (8 分)

Given a binary search tree T, you are supposed to output all the elements in the left subtree of X, in *ascending order*.

Format of functions:

```
void print_left_subtree( Tree T, int X );
```

Where Tree is defined as:

```
typedef struct TreeNode *Tree;

struct TreeNode {
    int element;
    Tree left;
    Tree right;
};
```

Sample program of judge:

```
#include <stdio.h>
#include <stdlib.h>

typedef struct TreeNode *Tree;

struct TreeNode {
    int element;
    Tree left;
    Tree right;
};

Tree build_tree(); /* details omitted */

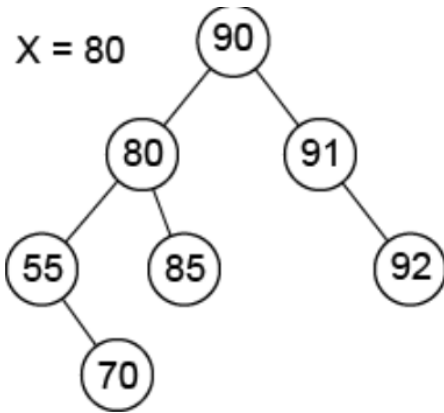
void print_left_subtree(Tree T, int X);

int main()
{
    Tree T;
    int X;
    T = build_tree();
    scanf("%d", &X);
```

```
print_left_subtree(T, X);
```

```
return 0; } /* Your function(s) will be put here */
```

Sample Output:



For the sample above, your program should output:55 70 end

For an empty tree, simply output:empty tree

In case X is not found in T , output:X not found

2021-2022 学年第一学期期末考试试卷参考答案

一、判断题 (10 题)

1、【正解】T

【解析】最大堆，又称大根堆（大顶堆）是指根结点（亦称为堆顶）的关键字是堆里所有结点关键字中最大者，属于二叉堆的两种形式之一。给定具有唯一关键字的最大堆，至少有 $d-1$ 关键字比第 d 层中的任何关键字都大（根级别为 1）

【考点延伸】堆

2、【正解】F

【解析】在用数组表示的循环队列中,front 值不一定小于等于 rear 值。在一个循环队列中，如果有若干个元素出队了，而且入队元素的总个数超过了数组的大小个数，是会发生 front 大于 rear 这种现象的。

【考点延伸】队列

3、【正解】T

【解析】假设中心轴是 28，那么 28 之前的数都比 28 小，28 后面的数都比 28 大。显然，题目中的一组数，不符合该情况。

【考点延伸】快速排序

4、【正解】T

【解析】若当前 key 与原来 key 产生相同的哈希地址，则当前 key 存在该地址后偏移量为(1,2,3...)的二次方地址处。

二次探测：每次加的步长随机或者是一定规律的数值。

再哈希法：哈希化有两个，第二个哈希化不能输出 0，不能与第一个哈希化相同。

第二个哈希化得到的结果是当前冲突时，索引需要加的值,即是在二次探测的基础上将步长的改进。

【考点延伸】哈希表

5、【正解】F

【解析】经路径压缩后的时间复杂度大约为 $O(|E|\log_2|E|+\alpha|V|)$

【考点延伸】最小生成树

6、【正解】F

【解析】两次排序的增量是 5 和 3

【考点延伸】希尔排序

7、【正解】T

【解析】该程序的主要语句是 sum++;所以时间复杂度是指该语句的执行次数。

【考点延伸】时间复杂度

8、【正解】T

【解析】在有 4 个连通分量的无向无环图 G 中，边的数目比顶点的数目少 4 个。

【考点延伸】连通分量

9、【正解】T

【解析】由树的节点数与度的关系知： $n_1+2n_2+3n_3+4n_4+1=n_0+n_1+n_2+n_3+n_4$,化简之后 $n_0=n_2+2n_3+3n_4+1, n_2+2n_3+3n_4+1-(n_2+n_3+n_4)=n_3+2n_4+1>0$,故 $n_2+n_3+n_4<n_0$

【考点延伸】树的度数

10、【正解】F

【解析】FirstChild-NextSibling 并没有说明一定要选最左边的作为 FirstChild,同样的，对于 Sibling 到底是哪一个点也是不在乎的。

【考点延伸】树

二、单选题(20 题)

1、【正解】C

【考点延伸】单源最短路径

2、【正解】B

【解析】完全二叉树中任何一层最左的节点编号 n ,则其左子树为 $2n$,右子树为 $2n+1$ 。

【考点延伸】完全二叉树

3、【正解】D

【解析】仅进行两次 partition 后，数组即可有序。

【考点延伸】排序

5、【正解】D

【解析】最早完成时间和最迟完成时间均为 14

【考点延伸】最早最迟完成时间

5、【正解】B

【解析】关节点是指在数据结构的某图中，如果删除顶点 V 以及 V 相关的边后，图的一个连通分量分割为两个或两个以上的连通分量，则称顶点 V 为该图的一个关节点。该图的关节点是：B C H G

【考点延伸】图的关节点

6、【正解】B

【解析】树与二叉树的转换依据的是“左孩子，右兄弟”的规则。以只有右子树的高度为 3 的二叉树为例，该二叉树的叶子结点为 1，转变成树之后叶子结点变为 3， $1<3$.故 B 是正确的。

【考点延伸】树与二叉树

7、【正解】B

【解析】二叉查找树 (Binary Search Tree)，(又：二叉搜索树，二叉排序树) 它或者是一棵空

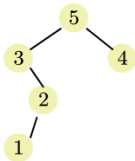
树，或者是具有下列性质的二叉树：若它的左子树不空，则左子树上所有结点的值均小于它的根结点的值；若它的右子树不空，则右子树上所有结点的值均大于它的根结点的值。由图可知，该二叉树的根节点只有右子树，故所有节点的值均应比根节点的值大，排除 C。以此类推可排除 AD。

【考点延伸】二叉搜索树

8、【正解】A

【解析】该二叉树为

【考点延伸】二叉树的遍历



9、【正解】D

【解析】选择排序，堆排序，快速排序可以在每次运行后找到排序列表中至少一个元素的最终位置

【考点延伸】排序

10、【正解】D

【解析】Starting from B: (B,D) -> (B,E) -> (A,B) -> (A,C) -> (D,F) -> (F,G)

【考点延伸】最小生成树

11、【正解】

12、【正解】D

【解析】 $h(92)=92\%13=1$; $h(81)=81\%13=3$; $h(58)=58\%13=6$ 。因为 58 是先插入到索引 6 的位置，所以索引 6 的位置为 58。

【考点延伸】哈希表

13、【正解】C

【解析】 $H(18)=18\%13=5$; $H(23)=23\%13=10$;

$H(4)=4\%13=4$; $H(26)=26\%13=0$; $H(50)=50\%13=11$; $H(33)=33\%13=7$; $H(17)=17\%13=4$; (冲突)。此时装载密度为 $6/13=0.46$

【考点延伸】哈希函数

14、【正解】D

【解析】该程序运行是的主要语句为 $i++$ ；又因为 while 中的语句是 3 次方增加的，因此时间复杂度为 D

【考点延伸】时间复杂度

15、【正解】A

【考点延伸】树的并查集

16、【正解】A

【解析】栈具有先进后出的性质。按照此性质计算即可

【考点延伸】栈

17、【正解】A

【解析】除了第一层根节点外，每层只有3个节点，此时树的高度最大为21层。 $h=20$

【考点延伸】树

18、【正解】D

【解析】当树为完全二叉树时，高度最小；当树的每个节点只有一颗子树时，高度最大；

【考点延伸】树的高度

19、【正解】B

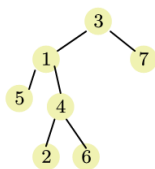
【解析】无向连通图有欧拉回路的充要条件是所有顶点都有偶数度

【考点延伸】欧拉回路

20、【正解】B

【解析】该二叉树为

【考点延伸】二叉树的遍历



三、程序填空题

1、【解析】 $\phi a[i][j] \neq 0 \quad \phi --Indegree[j]$

【考点延伸】拓扑排序算法

2、【解析】 $\phi child \neq N-1 \&\& A[Child] > A[Child+1] \quad \phi PercDown(A,0,i)$

【考点延伸】排序算法

四、fn 函数体

1、【解析】

```
void inorder(Tree T)
```

```
{
    if(!T)return;
    if(T->left)inorder(T->left);
    printf("%d ",T->element);
    if(T->right)inorder(T->right);
    return;
}
```

```
void print_left( Tree T, int X )
```

```
{
    if(!T){printf("x not found");return;}
    if(X<T->element){print_left(T->left,X);}
    else if(X>T->element){print_left(T->right,X);}
    else if(X==T->element){inorder(T->left);printf("end");}
```

```
        return;
    }

void print_left_subtree( Tree T, int X )
{
    if(!T)
    {
        printf("empty tree");
        return;
    }
    if(X<T->element){print_left(T->left,X);}
    else if(X>T->element){print_left(T->right,X);}
    else if(X==T->element){inorder(T->left);printf("end");}
    return;
}
```

【考点延伸】 二叉树算法